

GISWIL, Switzerland

WMO: 066570

Lat: 46.849N

Lon: 8.190E

Elev: 1558

StdP: 13.89

Time Zone: 1.00 (EUC)

Period: 11-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	16.3	20.5	10.3	9.9	16.7	16.5	13.4	23.0	18.4	48.6	15.1	46.8	3.9	80	0.462

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.4	86.3	69.3	82.9	68.2	79.5	66.4	71.0	82.9	69.3	80.1	67.7	77.6	3.5	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.3	106.5	75.5	65.5	99.9	73.4	64.2	95.1	71.7	35.8	83.5	34.3	80.2	33.0	77.9	77.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
14.6	10.7	7.7	DB	9.0	91.9	4.9	2.3	5.5	93.6	2.6	94.9	-0.2	96.2	-3.7	97.9
			WB	8.1	74.6	5.0	1.9	4.5	76.0	1.5	77.1	-1.3	78.1	-5.0	79.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	49.7	34.2	33.9	41.6	48.9	55.5	63.6	66.6	65.9	58.8	50.7	41.2	35.1	(6)
	DBStd	13.29	6.79	7.25	5.60	6.09	5.98	5.77	5.18	5.09	5.06	5.74	6.25	6.47	(7)
	HDD50	2114	492	452	264	90	18	0	0	0	2	62	272	462	(8)
	HDD65	5802	955	872	725	483	300	95	43	53	195	442	713	926	(9)
	CDD50	2020	1	0	4	57	187	407	513	492	264	85	9	1	(10)
	CDD65	234	0	0	0	0	4	52	91	79	8	0	0	0	(11)
	CDH74	2397	0	0	0	18	90	596	884	724	83	2	0	0	(12)
	CDH80	655	0	0	0	0	8	178	274	192	3	0	0	0	(13)
(14) Wind	WSAvg	2.5	2.9	2.6	2.7	2.8	2.7	2.5	2.4	2.2	2.0	1.9	2.3	2.8	(14)
(15) Precipitation	PrecAvg	61.5	3.7	3.9	4.1	4.6	6.5	6.7	7.0	6.8	4.8	4.2	4.4	4.5	(15)
	PrecMax	76.8	9.4	11.2	9.4	9.2	10.9	10.6	13.5	15.5	9.4	7.2	10.3	10.7	(16)
	PrecMin	47.6	0.3	1.3	1.7	1.3	1.5	3.6	2.8	1.6	2.1	0.2	0.0	0.2	(17)
	PrecStd	7.4	2.3	2.5	2.0	1.9	2.6	1.7	2.6	2.8	1.7	1.7	2.3	2.5	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	60.4	59.0	65.9	76.7	80.5	90.6	91.4	89.1	80.0	71.8	68.7	58.6	(19)
		MCWB	50.1	47.9	51.2	56.5	62.7	71.4	72.0	70.0	66.4	59.3	50.1	46.6	(20)
	2%	DB	52.2	53.1	62.2	72.1	76.4	85.1	86.5	85.1	76.6	67.0	59.4	52.4	(21)
		MCWB	45.7	44.4	49.5	55.4	62.0	69.3	69.1	69.5	65.1	57.7	51.5	46.7	(22)
	5%	DB	46.8	48.2	58.6	68.8	73.1	80.6	82.7	81.6	72.8	63.4	53.7	48.4	(23)
		MCWB	42.6	42.0	48.6	54.5	59.2	67.0	68.1	68.4	63.0	56.4	49.6	44.4	(24)
	10%	DB	43.5	43.8	54.2	63.8	69.0	76.7	78.8	78.0	69.3	60.1	50.0	44.6	(25)
		MCWB	40.7	40.2	46.1	52.3	57.8	64.5	66.1	66.8	61.4	55.6	47.3	41.9	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.3	49.6	53.3	59.1	65.7	73.2	73.6	72.7	67.6	61.0	53.8	49.5	(27)
		MCDB	58.6	56.2	63.0	73.3	78.9	86.8	88.3	83.2	76.3	66.7	59.5	55.0	(28)
	2%	WB	45.9	45.3	50.7	56.8	63.1	70.3	70.7	70.8	66.0	59.1	51.7	47.0	(29)
		MCDB	49.8	52.2	60.5	68.7	72.8	82.2	83.5	81.7	75.0	63.2	57.5	51.5	(30)
	5%	WB	43.4	42.6	48.7	54.6	60.8	68.2	68.8	69.2	64.0	57.6	49.8	44.4	(31)
		MCDB	46.4	47.3	56.7	66.1	69.5	79.3	79.9	79.3	70.7	61.4	54.7	47.8	(32)
	10%	WB	41.2	40.2	46.7	52.9	58.8	66.0	67.2	67.6	62.4	56.2	47.9	42.1	(33)
		MCDB	43.0	43.9	53.5	62.3	66.7	74.7	77.0	76.4	67.9	59.6	50.2	44.4	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	12.0	14.8	20.6	22.7	20.1	20.1	20.4	20.5	18.4	16.4	12.9	12.4	(35)
		MCDBR	17.9	24.6	29.9	33.1	28.9	28.8	28.8	26.9	24.2	22.8	20.1	17.0	(36)
	5% WB	MCWBR	12.5	15.6	18.4	17.5	14.8	14.3	13.5	13.3	13.5	14.3	12.4	11.4	(37)
		MCWBR	17.3	22.5	28.2	29.8	25.5	26.7	25.4	25.3	22.1	19.5	19.3	16.3	(38)
(40) Clear-Sky Solar Irradiance	taub		0.294	0.328	0.382	0.432	0.452	0.457	0.458	0.443	0.408	0.373	0.327	0.291	(40)
		taud	2.313	2.213	2.123	2.067	2.087	2.139	2.140	2.193	2.266	2.326	2.365	2.336	(41)
	Ebn at Noon		250	262	263	260	259	258	256	254	252	243	235	237	(42)
		Edh at Noon	24	33	42	49	49	47	47	43	36	29	23	21	(43)
(44) All-Sky Solar Radiation	RadAvg		442	729	1108	1396	1553	1821	1791	1558	1201	779	455	365	(44)
	RadStd		53	95	158	213	151	167	158	145	116	86	62	53	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (121 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	+1.30	+0.72	+0.56	N/A	N/A	+103	+29

Nomenclature: See separate page